Directional transfer of chemical-gene effects from rodents to humans: an independence-guarded meta-analysis of the Comparative Toxicogenomics Database

Abstract

Animal-to-human extrapolation is the central validity problem in toxicology, yet the reliability with which a chemical's directional effect on a gene transfers across species has not been quantified at scale. Using the Comparative Toxicogenomics Database, we assembled 1,165,406 curated chemical-gene-endpoint effects across human, mouse and rat and asked how often a rodent directional call (increases or decreases) matches the human call. The evidence base is curated in vivo and in vitro findings; exposure doses and durations are not standardized across the source literature, so the estimand is literature-level directional concordance rather than controlled biological replication. Because a naive comparison is circular when both calls come from the same publication, the primary analysis was restricted to 40,779 effects with disjoint PubMed identifiers across species. Independent directional agreement was 62.0% (95% CI 61.5-62.5), only 11.4 points above chance and close to the 66.2% mouse-versus-rat within-rodent benchmark; human-specific divergence was 4.2 points (95% CI -1.1 to 10.9). Shared-publication co-curation inflated agreement to 89.2%. Transfer varied more by molecular endpoint (activity 89%, expression 61%, methylation at chance) and replication depth

(61% at one report per species, rising to 100% at five) than by chemical class (variance component ≈1%), localising to the specific chemical-gene pair (intraclass correlation 0.44). Transfer was weakly predictable for unseen chemicals (grouped cross-validated AUC 0.58), and human-rodent protein identity was not associated with transfer (odds ratio 0.97 per 10 points, 95% CI 0.94-1.00). Cross-species directional disagreement is largely generic and context-dependent rather than tracking evolutionary distance.
Keywords: toxicogenomics, cross-species extrapolation, new approach methodologies, reproducibility, Comparative Toxicogenomics Database, ortholog conservation

1. Introduction

Almost everything we believe about how chemicals perturb human biology is inferred, at some remove, from rodents. That inference is the foundation of regulatory toxicology and of preclinical drug development, and it is also their best-documented weakness: across large bodies of comparisons, animal results translate to humans with wide and often disappointing concordance (Leenaars et al., 2019; Marshall et al., 2023), big safety databases confirm that animal findings are informative but far from sufficient predictors of human toxicity (Clark and Steger-Hartmann, 2018; Liu and Fan, 2026), and even at the level of the transcriptome the concordance between species has been contested strongly enough to support opposite published conclusions from the same data (Seok et al., 2013; Takao and Miyakawa, 2015). The response from regulators and funders has been to re-weight the evidence base: from the founding principles of replacement, reduction and refinement

(Russell and Burch, 1959) to the current pivot toward human-relevant new approach methodologies and the regulatory acceptance of non-animal evidence in drug submissions (Wu et al., 2026). All of this presupposes an answer to a question that has never been quantified at the level of individual molecular effects: *when a chemical moves a gene in a given direction in a rodent, how often does the same directional effect hold in a human?*

Individual studies have compared species for one chemical or one organ, but a comprehensive, effect-level answer requires a resource that has already harmonised the vocabulary of chemical action across species and the literature. The Comparative Toxicogenomics Database (CTD) is that resource: a manually curated compendium of chemical-gene, chemical-disease and gene-disease relationships drawn from the published literature, in which each chemical-gene interaction carries a signed action (for example, "increases expression" or "decreases activity"), an organism, and its supporting references (Mattingly et al., 2003; Davis et al., 2025). This makes it possible to line up, for a given chemical, gene and molecular endpoint, the directional call curated in humans against the call curated in rodents, across tens of thousands of independent instances.

The apparently obvious analysis, simply asking how often the human and rodent calls agree, is fatally circular. A large fraction of cross-species pairs are curated from the *same* publications: a single paper that reports an effect in both a human cell line and a mouse contributes a matched pair whose agreement is guaranteed by construction and reflects the curation pipeline rather than biology. Any honest estimate must therefore isolate the pairs whose human and rodent evidence come from disjoint references. This distinction between apparent and independent concordance is, to our knowledge, absent from prior cross-species toxicogenomic work, and it turns out to matter enormously.

We report an independence-guarded meta-analysis of directional transfer in CTD. We pre-specified four aims: (i) to estimate independent human-rodent directional agreement overall and stratified by molecular endpoint, chemical class and replication depth, benchmarked against a within-rodent (mouse-versus-rat) agreement benchmark; (ii) to decompose non-agreement into within-species inconsistency and genuine between-species divergence; (iii) to test whether transfer is predictable for chemicals never seen in training, using grouped cross-validation; and (iv) to test, against a pre-registered anchor lying entirely outside CTD curation, whether evolutionary conservation of the gene explains transfer. Framing throughout is deliberately descriptive: the substrate is curated published findings, and the estimands are conditional on curation.

2. Materials and methods

2.1. Data source and release

We used the CTD bulk data files (release of 30 July 2026), comprising the curated chemical-gene interaction file and the chemical, gene and pathway vocabularies (Davis et al., 2025). The external conservation anchor was built from NCBI `gene_orthologs` and Ensembl Compara, both independent of CTD (Brown et al., 2015; Austine-Orimoloye et al., 2025). All definitions below were fixed in a configuration file before analysis, and the full pipeline is deterministic given a fixed random seed.

2.2. *Experimental system and human relevance*

The experimental systems underlying this study are the curated in vivo (human, mouse and rat) and in vitro chemical-gene findings recorded in CTD; the analysis operates on directional calls abstracted from many thousands of primary studies rather than on a single controlled exposure. Human relevance is intrinsic to the design rather than extrapolated: the primary estimand compares each rodent directional effect with the *directly curated human* directional effect for the same chemical and gene, so the human comparator is measured, not inferred from a model species. Because the substrate is aggregated curated literature, the exposure dose or concentration and duration, and the CAS number, source and purity of each chemical, are properties of the individual source experiments and are neither standardized nor controllable at the level of this meta-analysis; we therefore treat them as uncontrolled and report agreement conditional on curation, and we do not draw dose- or purity-dependent conclusions. Randomization of treatment groups does not apply to a retrospective curated census; in its place, unit-level replication is the number of independent supporting publications per species (the replication-depth stratification of Section 3), and all inferential procedures (Wilson intervals, chemical-cluster bootstraps and grouped cross-validation) are specified below.

2.3. *Effect keys and directional consensus*

The unit of analysis is an *effect key*: a (chemical, gene, molecular endpoint) triple. We retained single-action, signed interactions whose action carried an interpretable direction, that is, "increases" or "decreases" applied to one of the molecular endpoints expression, activity, methylation, phosphorylation, secretion, abundance, cleavage, metabolic processing, stability, uptake, transport and related aspects; multi-action and unsigned interactions were excluded. Records were grouped into three species by NCBI taxonomy identifier: human (9606), mouse (10090) and rat (10116); mouse and rat were pooled as "rodent" for the primary contrast and analysed separately for the within-rodent (mouse-versus-rat) benchmark.

For each effect key and species we counted "increases" and "decreases" curations and assigned a consensus direction: $+1$ if only increases were curated, -1 if only decreases, and 0 (ambiguous) if both directions or none appeared. We also recorded, per species, the set of supporting PubMed identifiers. The analytic table contained 1,165,406 effect keys.

Two points about interpretation follow from this construction, and we return to them throughout. First, a matched human and rodent call need not derive from the same tissue, dose, exposure duration or cell type; what we measure is therefore the concordance of curated directional calls across the literature, which is a necessary condition for biological reproducibility but is not a controlled biological replication of it. Second, every estimand is conditional on what has been published and curated, and inherits the corresponding selection. We use "agreement" and "transfer" throughout as shorthand for this literature-level directional concordance, not as a direct claim about biological reproducibility.

2.4. Independence guard

For every key measured in both human and rodent we computed the intersection of their supporting PubMed identifiers. A pair was *independent*

(PMID-disjoint) if that intersection was empty, so that the human and rodent directional calls rest on non-overlapping primary literature. The primary population comprised keys measured in both species, with an unambiguous consensus in each, and with disjoint supporting references. Keys sharing at least one reference were analysed separately to quantify co-curation inflation.

2.5. Aim 1: reproducibility

The primary estimand was the probability that the human consensus direction equals the rodent consensus direction, among independent unambiguous pairs. We report the point estimate with a Wilson interval (Wilson, 1927) and, because keys are nested within chemicals and genes, a cluster (block) bootstrap with chemicals as the resampled unit and 1,000 replications (Davison and Hinkley, 1997). A chance-agreement baseline was computed from the marginal direction frequencies in each stratum. Mouse-versus-rat agreement, estimated identically, provided a within-rodent benchmark: an upper reference for the agreement attainable between two closely related mammals studied through the same curated literature. This benchmark is not a pure reproducibility floor, because mouse and rat are distinct species with genuine biological differences; it bounds the combination of curation reproducibility and closely-related-species divergence. Because it already absorbs some true biological divergence, any excess disagreement for human-versus-rodent over this benchmark is a conservative estimate of the specifically human-versus-rodent component. Pre-specified strata were molecular endpoint, coarse chemical class from the MeSH chemical tree, and replication depth (the minimum number of supporting references on either side).

2.6. Aim 2: decomposition

Non-agreement can arise two ways: a species may be internally inconsistent (the same key curated both up and down within one species, reflecting context, time or tissue dependence, or curation noise), or each species may be internally consistent yet disagree with the other. We measured within-species inconsistency as the fraction of multiply-curated keys that were internally ambiguous, separately in each species. We defined human-specific divergence as the difference between human-versus-rodent disagreement and the mouse-versus-rat benchmark, with a cluster bootstrap interval; a value near zero indicates that cross-species disagreement is already present within a single clade rather than being a specifically human-versus-rodent phenomenon. Finally, we partitioned the variance of the binary agreement outcome across grouping factors (chemical, gene, endpoint, chemical class, and the chemical-gene pair) using a one-way intraclass correlation coefficient (Shrout and Fleiss, 1979), which estimates the share of agreement variance attributable to each factor.

2.7. Aim 3: predictability

We asked whether, given only rodent-side information, the human direction is predictable for a chemical never seen during training. The outcome was agreement (human direction equal to rodent direction) on the independent unambiguous set. Features were known at prediction time and used no human-side information: molecular endpoint, chemical class, rodent consensus direction, rodent evidence depth and support, and chemical and gene promiscuity (the number of distinct keys involving each) and gene pathway membership. We fitted a regularised logistic model and a histogram gradient-boosting classifier (Friedman, 2001; Ke et al., 2017; Pedregosa et al., 2011) under *grouped* five-fold cross-validation with chemicals as groups, so that no chemical appeared in both training and test folds; this prevents the optimistic leakage that ordinary cross-validation would introduce through repeated chemicals (Kaufman et al., 2012). We report discrimination (area under the ROC curve, average precision), calibration (reliability curve, Brier score) and model-agnostic permutation importance, together with a leave-one-chemical-class-out generalisation test.

2.8. Aim 4: external conservation anchor

Every quantity above is internal to CTD curation. To test whether the result is self-referential, we pre-registered, before any Aim 4 modelling, a single external anchor: evolutionary conservation of the human gene to its rodent ortholog, taken from sources outside CTD. Graded conservation was the protein percent identity of the human gene to its mouse and rat orthologs from Ensembl Compara (we used the per-gene mean and minimum across the two rodents) (Austine-Orimoloye et al., 2025); orthology status was the one-to-one flag from Ensembl, with the NCBI `gene_orthologs` one-to-one flag as a secondary anchor (Brown et al., 2015). The directional, pre-specified hypothesis was that transfer increases with conservation: if cross-species failure were driven by molecular divergence, effects on highly conserved genes should transfer more reliably. We related agreement to conservation quartile with cluster-bootstrap intervals, estimated a chemical-cluster-robust logistic trend (odds ratio per 10 percentage points of identity), contrasted one-to-one against other orthology, and tested whether conservation adds discrimination to the Aim 3 model under the same grouped cross-validation. The relevance of orthology conjecture debates to functional transfer motivated this design

(Nehrt et al., 2011; Chen and Zhang, 2012).

2.9. Software and reproducibility

Analyses used Python with pandas, NumPy, SciPy, statsmodels and scikit-learn (Pedregosa et al., 2011). Cluster-robust inference used chemical as the clustering unit throughout. Code and non-identifiable aggregate tables are openly available at `https://github.com/january-msemakweli/Cross-Species-Directional-Transfer-Chemical-Gene-Effects`.

3. Results

3.1. Independent cross-species agreement is modest, and co-curation inflates it

Of 1,165,406 effect keys, 61,439 were curated in both a human and a rodent context. Restricting to keys with an unambiguous consensus in each species and, critically, to those whose human and rodent evidence came from disjoint publications left 40,779 independent pairs. Among these, the human and rodent directional calls agreed 62.0% of the time (95% CI 61.5-62.5; cluster bootstrap 58.7-65.4), which is 11.4 percentage points above the 50.6% chance baseline (Table 1, Figure 1). When the independence guard was removed and pairs sharing at least one reference were examined, apparent agreement rose to 89.2% (88.3-90.1). The gap between these two numbers is the central methodological result: roughly two-thirds of the naive concordance one would report from CTD without an independence guard is an artefact of the same paper being curated into both species.

The independent estimate is best read against a within-rodent benchmark. Mouse-versus-rat agreement, estimated identically on 25,615 pairs, was 66.2% (65.6-66.7). This is the agreement attainable between two closely related rodents through the same curated literature; it blends curation reproducibility with any genuine mouse-rat biological differences, and so is an upper reference rather than a pure noise floor. Independent human-rodent agreement (62.0%) sits only just below it, which already reframes the problem: most of what looks like a human-specific species barrier is present even between two rodents.

Table 1: Directional agreement of chemical-gene effects across species. The primary estimate is restricted to pairs whose human and rodent evidence come from disjoint publications; the shared-publication stratum shows the inflation that an independence guard removes; the mouse-versus-rat stratum is a within-rodent benchmark that blends curation reproducibility with genuine mouse-rat biological differences. Bootstrap intervals resample chemicals.

Comparison	Pairs (n)	Agreement	Wilson 95% CI[a]	Bootstrap 95% CI[b]	Chance	Excess[c]
Human vs rodent, independent (primary)	40,779	**62.0%**	61.5-62.5	58.7-65.4	50.6%	+11.4
Human vs rodent, all pairs	45,557	64.9%	64.4-65.3	61.7-68.4	50.9%	+14.0
Human vs rodent, shared publication	4,778	89.2%	88.3-90.1	80.2-96.1	54.5%	+34.8
Mouse vs rat (within-rodent benchmark)	25,615	66.2%	65.6-66.7	58.7-74.8	45.2%	+21.0

[a] Wilson score interval for the point estimate, treating pairs as independent.

[b] Cluster bootstrap resampling chemicals (1,000 replications); wider than the Wilson interval because it propagates within-chemical correlation. This is the interval used for all inference and shown in the figures.

[c] Percentage points above the stratum-specific chance baseline derived from marginal direction frequencies.

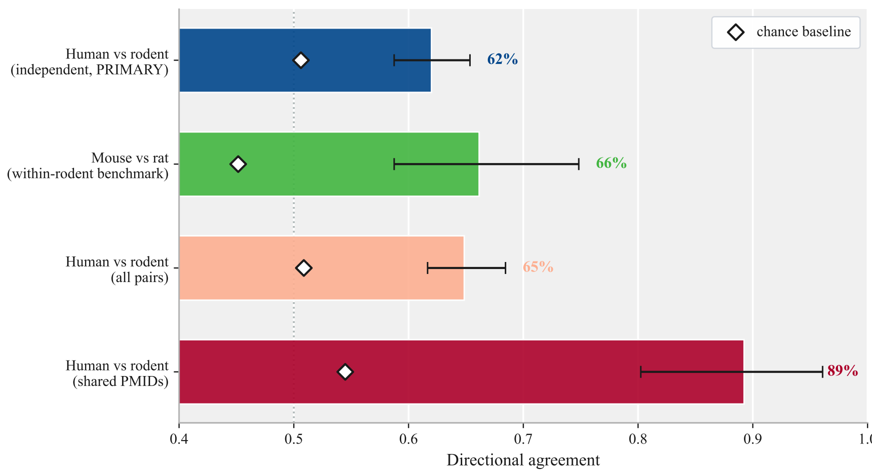

Figure 1: Directional agreement of chemical-gene effects by comparison. Bars are point estimates with cluster-bootstrap 95% intervals (chemicals resampled); diamonds mark the stratum-specific chance baseline. Independent human-rodent agreement (navy) is far below the shared-publication estimate and sits just below the mouse-versus-rat within-rodent benchmark.

3.2. Transfer is governed by molecular endpoint and replication depth, not chemical class

Agreement was strongly heterogeneous across molecular endpoints (Figure 2, Supplementary Table S1). Functional endpoints transferred well: cleavage 96.7% (n=182), activity 89.3% (n=1,106), secretion 85.7% (n=286) and phosphorylation 81.5% (n=744). Bulk molecular endpoints transferred poorly: gene expression, which dominates the data (n=36,986), agreed only 60.8% of the time, and DNA methylation (n=1,102) agreed 41.9%, essentially at its 40.9% chance baseline. A directional effect on methylation curated in a rodent therefore carries almost no information about the direction in a human.

Replication depth was the strongest ordered signal (Figure 3, Supplementary Table S2). Agreement rose monotonically from 60.7% for effects supported by a single report on the thinner side, to 76.5% at two, 90.6% at three, 95.6% at four, and 100% at five or more supporting reports per species. Multiply-replicated directional effects transfer nearly deterministically; the modest headline number is a property of the long tail of singly-curated effects, not of the well-established ones.

By contrast, chemical class barely mattered. Agreement ranged only from 59.1% for metals to 68.6% for heterocyclic organics (Supplementary Table S3), a spread small enough that, as shown next, chemical class explains almost none of the variance in transfer.

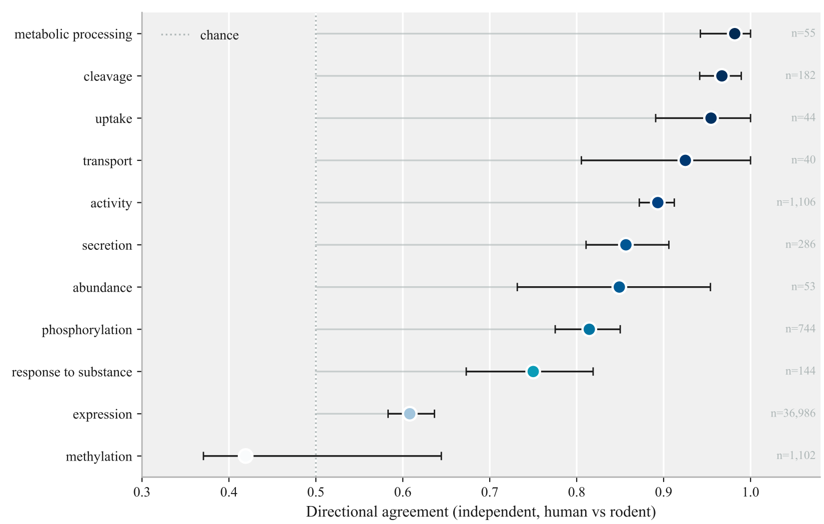

Figure 2: Independent human-rodent directional agreement by molecular endpoint, ordered by agreement, with cluster-bootstrap 95% intervals and per-stratum sample sizes. The dotted line marks chance. Functional endpoints (activity, secretion, phosphorylation, cleavage) transfer well; bulk expression is modest and methylation sits at chance.

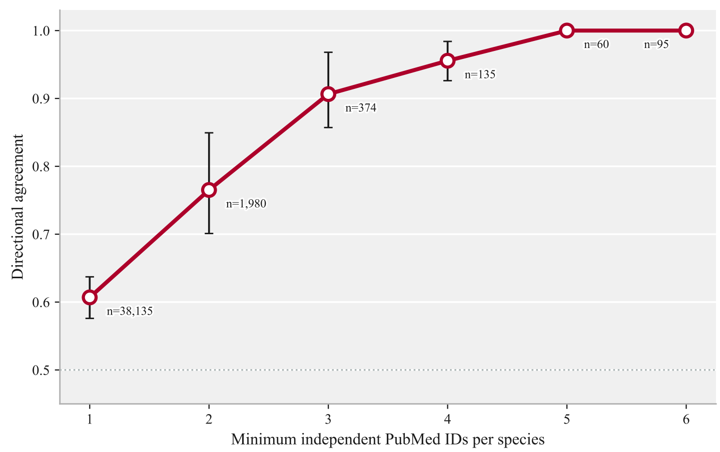

Figure 3: Independent directional agreement as a function of replication depth, the minimum number of supporting publications per species, with cluster-bootstrap 95% intervals. Agreement climbs from ∼61% at a single report to 100% at five or more.

3.3. Cross-species disagreement is largely generic and localises to the chemical-gene pair

Two analyses locate the disagreement. First, within-species inconsistency is large: of keys curated more than once, 58.3% were internally ambiguous in humans, 54.5% in mouse and 65.9% in rat (Supplementary Table S4). A substantial share of apparent cross-species disagreement is simply that a single species does not reproduce its own directional call. Second, human-specific divergence is small: human-versus-rodent disagreement (38.0%) exceeded the mouse-versus-rat benchmark (33.8%) by only 4.2 percentage points, with a cluster-bootstrap interval that crosses zero (−1.1 to 10.9; Figure 4A). The extra penalty for crossing from rodent to human, over and above the penalty for crossing between two rodents, is not distinguishable from zero.

The variance decomposition (Figure 4B, Supplementary Table S5) is consistent. The intraclass correlation of the agreement outcome was 0.44 for the chemical-gene pair, but only 0.12 for molecular endpoint, 0.068 for gene, 0.046 for chemical and 0.005 for chemical class. Transferability is thus overwhelmingly a property of the specific chemical-gene combination, not of broad chemical categories: knowing the class of a chemical tells you almost nothing about whether its effects transfer, whereas the identity of the exact pair tells you a great deal.

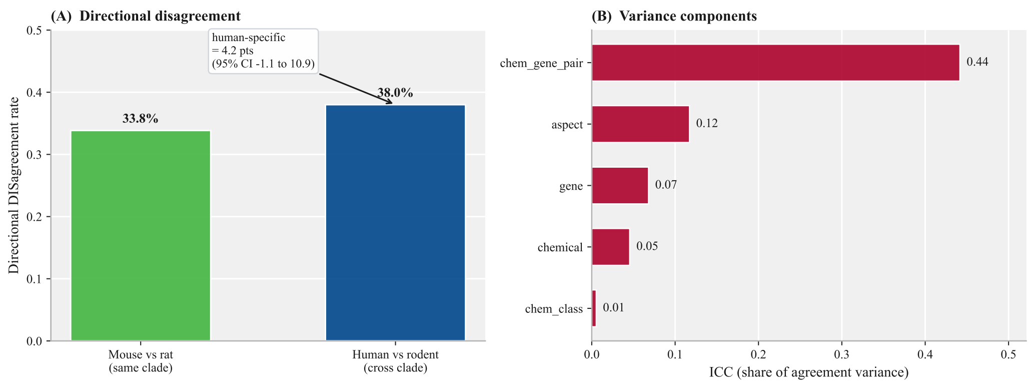

Figure 4: Decomposition of non-agreement. (A) Directional disagreement between mouse and rat (same clade) versus between human and rodent (cross clade); the human-specific excess is 4.2 percentage points with a bootstrap interval that includes zero. (B) Intraclass correlation of the binary agreement outcome by grouping factor; transfer localises to the specific chemical-gene pair, not to chemical class.

3.4. Transfer is only weakly predictable for unseen chemicals

Under grouped cross-validation with chemicals held out, rodent-side features predicted human agreement only weakly: the gradient-boosting model reached an area under the ROC curve of 0.58 (SD 0.02 across folds) and the logistic model 0.56, against a 62% prevalence baseline (Figure 5, Supplementary Table S6). The model was well calibrated (Figure 5A) and the most informative features were the rodent direction itself, gene promiscuity, molecular endpoint and rodent evidence depth (Figure 5B); chemical class contributed nothing, consistent with the variance decomposition. Leave-one-chemical-class-out AUCs ranged from 0.51 to 0.61. The honest reading is that no compact set of chemical- or gene-level covariates recovers transferability for a genuinely new chemical; the information lives at the pair level, which is by definition unavailable for unseen chemicals.

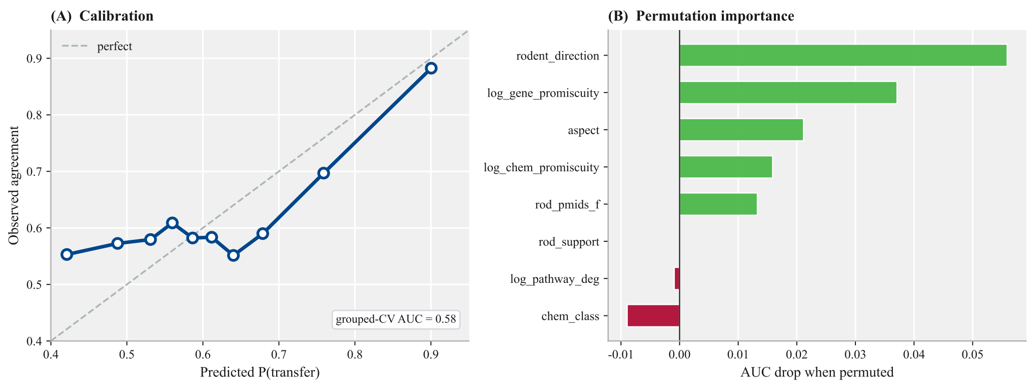

Figure 5: Predicting directional transfer for unseen chemicals under grouped (by chemical) five-fold cross-validation. (A) Reliability curve for the gradient-boosting model against the line of perfect calibration, with the grouped cross-validated AUC. (B) Permutation importance; the rodent direction, gene promiscuity, molecular endpoint and rodent evidence depth carry the signal, while chemical class does not.

3.5. Sequence conservation is not associated with transfer

The external anchor closes the circularity concern and returns a clean null. Conservation coverage was near-complete: 39,933 of 40,779 primary keys (97.9%) had an Ensembl human-rodent identity value (median 87.5%). Directional agreement was flat across the entire range of conservation (Figure 6A): 63.0% in the least-conserved quartile (median 70% identity), 61.8%

and 62.9% in the middle quartiles, and 60.5% in the most-conserved quartile (median 97% identity). The chemical-cluster-robust logistic trend was null, and if anything slightly negative: odds ratio 0.97 per 10 percentage points of identity (95% CI 0.94-1.00, $p = 0.07$). One-to-one orthology did nothing (Ensembl odds ratio 1.02, 95% CI 0.96-1.09; NCBI 0.92, 0.78-1.09; Figure 6B), and adding conservation to the Aim 3 model moved cross-validated AUC by 0.002 (Supplementary Table S8). How similar a gene's protein is between human and rodent was not associated with whether a chemical moves it in the same direction across species.

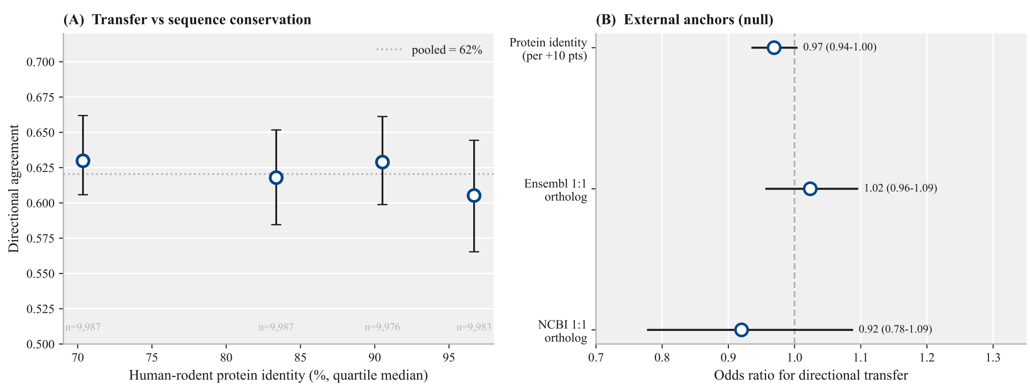

Figure 6: External conservation anchor (pre-registered). (A) Independent directional agreement by quartile of human-rodent protein identity, with cluster-bootstrap 95% intervals; agreement is flat across the full identity range. (B) Odds ratios for transfer from graded identity and from one-to-one orthology (Ensembl and NCBI); all straddle 1. Sequence conservation is not associated with directional transfer.

3.6. A catalogue of independent cross-species conflicts

The independence guard also yields a practical by-product: effects that are independently and repeatedly curated in opposite directions in humans and rodents. These are candidate true species divergences or curation errors worth targeted review, and they are enriched for well-studied environmental chemicals (for example bisphenol A, dioxin and paraquat) acting on stress-response and ribosomal genes. The full catalogue is provided in Supplementary Table S9.

4. Discussion

Using more than a million curated chemical-gene effects, we find that when a chemical's directional effect on a gene is established independently in rodents, it holds in humans about three fifths of the time, only modestly above chance and barely better than one rodent predicts another. The result is easy to misread in either direction, and the guardrails matter. Without an independence guard the same data yield 89% agreement, a number that would license comfortable extrapolation and is almost entirely an artefact of the same publication being curated into both species. With the guard, and against a within-rodent benchmark, the honest picture emerges: cross-species disagreement among curated directional chemical-gene effects looks far more like a problem of reproducibility and context dependence than a problem of phylogenetic distance. We are careful to state this as a claim about curated directional concordance, not about controlled biological reproducibility, which our data cannot measure directly.

Three findings support that reading and each is, we think, individually useful. First, the disagreement is largely generic rather than human-specific: the excess disagreement incurred by crossing from rodent to human, beyond that incurred by crossing between two rodents, is not distinguishable from zero, and a large share of disagreement is simply that a single species does not reproduce its own directional call. Second, what transfers is governed by *what is measured* and *how well it is established*, not by the kind of chemical. Functional endpoints such as enzyme activity and secretion transfer well; bulk expression is mediocre and DNA methylation is uninformative; and any effect replicated across a handful of studies transfers nearly deterministically. Third, an external and pre-registered axis of molecular similarity, protein sequence conservation, was not associated with transfer. We are cautious about causal language, but this is the pattern expected if diverged proteins were not the main driver: had the barrier been largely about sequence divergence, conservation should have tracked agreement, and it did not. The finding sits comfortably beside long-standing scepticism about the ortholog conjecture, the assumption that orthologous genes retain equivalent function across species (Nehrt et al., 2011; Chen and Zhang, 2012).

For practice, the actionable content is specific. A rodent directional call on a functional endpoint, or one supported by several independent studies, is a reasonable basis for a human expectation; a single-study call on expression or methylation is not. Because transferability localises to the specific chemical-gene pair and cannot be recovered from chemical class or from broad gene properties, screening strategies that triage chemicals by class, or that lean on sequence conservation to argue for cross-species relevance, are weakly grounded for directional prediction. These conclusions bear directly on the new approach methodologies agenda (Wu et al., 2026; Marshall et al., 2023): they argue for weighting evidence by endpoint type and replication rather than by taxonomic proximity, and they supply a quantitative baseline against which human-relevant assays can be judged.

The work also carries a general methodological warning for anyone mining curated biomedical databases for cross-species or cross-context concordance. When two facts are curated from a shared literature, their agreement is not evidence; it is bookkeeping. The 27-point gap between guarded and unguarded estimates here is a concrete measure of how badly that can mislead, and the same guard, disjoint supporting references, is applicable to any database that records provenance.

4.1. Limitations

The most important limitation is intrinsic to the substrate and to the estimand. CTD is curated from published findings, so what we measure is agreement between curated directional calls, not a controlled biological replication: a matched human and rodent call may come from different tissues, doses, exposure durations or cell types. Our estimates are therefore literature-level directional concordance, a necessary condition for biological reproducibility rather than a direct measurement of it, and every estimand is conditional on curation and inherits publication and curation bias, with effects measured in both species enriched for well-studied, "interesting" genes. This constraint cannot be removed by any CTD-only analysis, which is precisely why we anchored the study to an external axis; that the anchor returned a null strengthens rather than weakens the reproducibility reading, but it does not convert the design into an experimental one. Within-species "inconsistency" conflates genuine context dependence, real tissue, dose and time effects, with curation noise, and our data cannot fully separate them. The mouse-versus-rat benchmark is likewise not a pure noise floor: mouse and rat are distinct species, so it blends curation reproducibility with genuine mouse-rat biological divergence. Because it absorbs some real divergence, it slightly overstates the achievable reproducibility and so makes our estimate of the human-versus-rodent excess conservative rather than inflated. The pooling of mouse and rat into a single rodent group, while appropriate for the primary translational question, could mask species-specific patterns, although the mouse-versus-rat analysis argues against large ones. The predictive ceiling we report is a statement about compact, prediction-time features and does not exclude that richer molecular descriptors, unavailable here, would do better. Finally, our conservation anchor is protein sequence identity and one-to-one orthology; conservation of regulatory context or of expression, which we did not measure, could in principle behave differently, though there is no obvious mechanism by which it would rescue directional transfer with which sequence conservation was not associated.

5. Conclusions

Independent directional agreement between rodent and human chemical-gene effects is modest, is inflated roughly to 90% by shared-publication co-curation if the independence guard is dropped, and is barely above the agreement achievable between two rodents. Transfer is governed by molecular endpoint and replication depth rather than by chemical class, localises to the specific chemical-gene pair, is only weakly predictable for unseen chemicals, and is not associated with protein sequence conservation. Cross-species concordance of curated directional chemical-gene effects is better described as a problem of reproducibility and context dependence than of phylogenetic distance, and evidence should be weighted by what was measured and how firmly it was established, not by taxonomic proximity or sequence similarity. Because our estimands are literature-level directional concordance conditional on curation, they bound rather than directly measure biological reproducibility.

CRediT authorship contribution statement

The CRediT author contribution statement is provided on the separate title page to preserve double-blind review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The Comparative Toxicogenomics Database is openly available at `https://ctdbase.org`; NCBI `gene_orthologs` and Ensembl Compara are openly available from their respective providers. All analysis code and the non-identifiable aggregate tables and figures supporting this study are openly available at `https://github.com/january-msemakweli/Cross-Species-Directional-Transfer-Chemical-Gene-Effects` (and the corresponding

Zenodo archive linked from that repository's Releases). No restricted or patient-level data were used.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Grammarly AI in order to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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